

Namhoon Kim

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Education

Georgia Institute of Technology

Aug 2024 - present

Ph.D. Student in Machine Learning

- **Advisor:** [Sara Fridovich-Keil](#)
- **Research Interest:** Computational Imaging, 3D Reconstruction, Uncertainty Quantification

Yonsei University

Mar 2019 - Aug 2024

B.S. in Electrical and Electronic Engineering

- **GPA:** 3.94/4.3, *Magna Cum Laude* (Top 3 % in Engineering School), **Major GPA:** 4.13/4.3
- **Senior Thesis** (Advisor: [Seung Ah Lee](#)) End-to-End, Physics-Informed Lensless Segmentation Network.

Publications

Conferences

- S. Jeong **N. Kim**, and S. Fridovich-Keil, “Revisiting Shearlet Thresholding for 3D Denoising,” *NeurIPS* 2026 (under review).
- D. Bao, L. Wang, **N. Kim**, S. Fridovich-Keil, and J. Romberg, “Spectral Bias Often Emerges from Eigenspace Structure in Neural Network Training,” *NeurIPS* 2026 (under review).
- G. Jeong, S. Cho, M. Shon, **N. Kim**, K. Kim, W. Hwang, K. Seo, K. Kim, W. Kim, D. Ha, P. Venkatesan, K. Youn, R. Cherukuri, Y. Wang, S. Datta, A. Khan, and S. Yu “Physics-informed AI Accelerated Retention Analysis of Ferroelectric Vertical NAND: From Day-Scale TCAD to Second-Scale Surrogate Model,” *ICMC*, 2026.
- **N. Kim***, N. Moeini*, J. Romberg, and S. Fridovich-Keil, “3D Field of Junctions: A Noise-Robust, Training-Free Structural Prior for Volumetric Inverse Problems,” *ECCV*, 2026.
- **N. Kim** and S. Fridovich-Keil, “Grids Often Outperform Implicit Neural Representation at Compressing Dense Signals,” *NeurIPS*, 2025.
- **N. Kim** and S. Fridovich-Keil, “Towards Distribution-Shift Uncertainty Estimation for Inverse Problems with Generative Priors,” *IEEE CAMSAP* (invited paper), 2025.
- K. Lee, J. Bae, **N. Kim**, N. Beak, T. Kim, W. Park and S. Lee, “Differentiable Lensless Camera Design with Extended Field of View,” *Photonics Technology (at Optics and Photonics Congress)*, 2023.
- N. Baek, K. Lee, J. Bae, J. Park, T. Kim, D. Bae, **N. Kim**, W. Park and S. Lee, “Single-shot fabrication of light modulating phase mask for lensless camera using off-the-shelf components,” *Photonics Technology (at Optics and Photonics Congress)*, 2023.

Journals

- S. Lee, **N. Kim**, and S. Yoo, “Design-Space Analysis of Plug-and-Play Diffusion for PDE-Constrained Inverse Problems,” *TMLR*, 2026 (under review).
- **N. Kim**, A. Pananjady, A. Pourmorteza and S. Fridovich-Keil, “Perfusion Imaging and Single Material Reconstruction in Polychromatic Photon Counting CT,” *Medical Physics*, 2025 (under review).

Workshops

- **N. Kim***, N. Moeini*, J. Romberg, and S. Fridovich-Keil, “3D Field of Junctions: A Noise-Robust, Training-Free Structural Prior for Low-Dose CT,” *4th workshop on Vision-based Industrial InspectiON (VISION) (at CVPR)*, 2026.
- **N. Kim** and S. Fridovich-Keil, “Uncertainty Quantification for Inverse Problems with Generative Priors under Distribution Shift,” *Statistical Frontiers in LLMs and Foundation Models (at NeurIPS)*, 2024.

Selected Honors and Awards

- Next-generation Science and Technology Leader NET, Korean Federation of Science and Technology Societies** 2025
- Awarded \$2,000 research grant from KOFST supported interdisciplinary research projects.
“Efficient Management of Data Centers for Extreme Environments using Digital Twin Technology”
- 2nd prize, ML@GT Student Conference Best Poster Award:** Awarded \$200 Sep 2025
- 2nd prize, Mobility SW Hackathon, Hyundai Mobis** Feb 2023
- Developed an [In-vehicle Noise Resistant Voice Recognition model](#) with deep neural network.
 - Awarded \$4,000 for the prize and registered as an excellent talented human resource in Hyundai.
- 3rd prize, Yonsei Dream Library: Metaverse Content Production Contest** Feb 2022
- Awarded \$800 for prize for developing a [metaverse virtual university library](#) using Lua and Roblox IDE.
 - Established a server-client communication enabling the construction of a competitive studying system based on the exchange of data between users.

Mentoring

Sungwon Jeong, M.S. student at Georgia Institute of Technology Fall 2025 - present

Scholarships

Yonsei Eminence Scholarship, Yonsei University Fall 2022 - Spring 2024

- Awarded for academic excellence:
\$3000 (2/3 of Tuition) at Spring 2023, Fall 2023; \$2000 at Spring 2024; \$1000 at Fall 2022

Teaching Experiences

- Peer Tutor of course Digital Signal Processing (EEE4420), Yonsei University** Winter 2024
- Provided 16 hours of tutoring for 10 undergraduate students
- Residential College Program: Arduino Programming & Operation, Yonsei University** Spring 2023
- Appointed as a tutor and assisted 30 hours of lecture for 60 freshmen
- Social Engagement: Yon-In High School Students Mentoring, Yonsei University** Fall 2019
- Mentored high school students during the semester
- Peer Tutor of course Electricity & Magnetism(1) (EEE2030), Yonsei University** Summer 2022
- Provided 16 hours of tutoring for 40 undergraduate students

Leadership and Military Service

Republic of Korea Defense Intelligence Command (Military Service) Jul 2020 – Jan 2022
Army Sergeant, honorable discharge

YAI, Artificial Intelligence Club Yonsei University — President Dec 2023 – Jun 2024

Roboin, Robotics Club Yonsei University — Executive Staff Mar 2023 – Aug 2023

Jamsin High School — Student Vice President 2014

Skills

Programming: Python, MATLAB, C/C++, C#, Lua, Verilog

Software and Tools: PyTorch, Linux, HTML/JS/CSS, Latex, Unity, CAD (Blender, Fusion 360, PTC Creo)

Experiment: Optical systems setup, Arduino, Raspberry Pi, 3D Printing (Form3+, Ender)

Soft Skills: Time management, Leadership, Teamwork